DISCRETE STRUCTURES

CMSC250

Chapter 11: Number Theory and Divisibility

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Integers, divisibility, primes, and rigorous proofs



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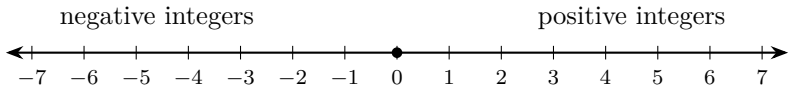
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1 Why number theory begins with integers

Number theory studies the structure of integers: multiples, divisors, primes, and remainders. Many statements that look obvious from examples require a formal argument for *every* integer. The most important habit is to turn each verbal property into an algebraic definition, then finish the proof by returning to that definition.

2 The integer number line

The integers continue without end in both directions. A number is positive when it lies to the right of zero and negative when it lies to the left. Zero is neither positive nor negative.

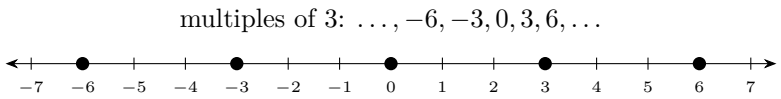


The set $\mathbb{Z}$ is closed under addition, subtraction, and multiplication. For example, if $r, s \in \mathbb{Z}$, then $3r - 2s \in \mathbb{Z}$. Integers are *not* closed under division: $1/2$ is not an integer.

3 Divisibility: definition and direction

For integers a, b with $a \neq 0$, the statement $a \mid b$ (read "a divides b") means that *there exists* an integer k such that $b = ak$. The number a is the divisor, b is the multiple, and k is an integer witness.

Key idea. $a \mid b$ means $b = ak$, not $a = bk$. For instance $3 \mid 12$ because $12 = 3 \cdot 4$; but $12 \nmid 3$.



The dots mark the multiples of 3. Notice that 0 is a multiple of every nonzero integer because $0 = a \cdot 0$.

Worked example: checking divisibility using the definition

Question. Does $-4 \mid 28$? Does $6 \mid 20$?

Solution. Yes, $-4 \mid 28$ because $28 = (-4)(-7)$ and $-7 \in \mathbb{Z}$. No, $6 \nmid 20$: an integer k with $20 = 6k$ would have to be $k = 10/3$, which is not an integer.

Worked example: a complete proof with integer witnesses

Question. For all integers a, b, d, with $d \neq 0$, if $d \mid a$ and $d \mid b$, then $d \mid (2a - 3b)$.

Proof. Let $a, b, d \in \mathbb{Z}$ be arbitrary, with $d \neq 0$. Assume $d \mid a$ and $d \mid b$. We will show $d \mid (2a - 3b)$. By the definition of divisibility, there exist $r, s \in \mathbb{Z}$ such that $a = dr$ and $b = ds$. Hence

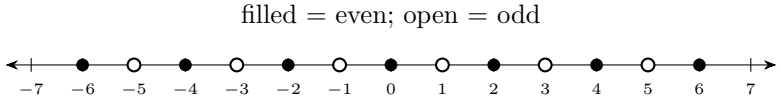
$$2a - 3b = 2dr - 3ds = d(2r - 3s).$$

Because $r, s \in \mathbb{Z}$ and integers are closed under multiplication and subtraction, $2r - 3s \in \mathbb{Z}$. Therefore $d \mid (2a-3b)$ by the definition of divisibility. Since the integers were arbitrary, the claim holds universally. ■

Why the proof works. We did not merely factor out d. We identified the integer witness $2r - 3s$ and explained why it is an integer.

4 Even and odd integers on the number line

An integer is **even** if it has the form $2k$ for some $k \in \mathbb{Z}$, and **odd** if it has the form $2k + 1$ for some $k \in \mathbb{Z}$. Consecutive integers alternate parity.



Worked example: odd plus odd is even

Question. Prove that the sum of two odd integers is even.

Proof. Let $a, b \in \mathbb{Z}$ be arbitrary odd integers. By the definition of odd, $a = 2r + 1$ and $b = 2s + 1$ for some integers r, s. Then

$$a + b = (2r + 1) + (2s + 1) = 2(r + s + 1).$$

By closure of $\mathbb{Z}$ under addition, $r + s + 1 \in \mathbb{Z}$. Thus $a + b$ is twice an integer, so $a + b$ is even by definition. The choice of a, b was arbitrary. ■

Worked example: the square of an odd integer is odd

Question. Prove that for every odd integer n, n^2 is odd.

Proof. Let $n \in \mathbb{Z}$ be an arbitrary odd integer. Write $n = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

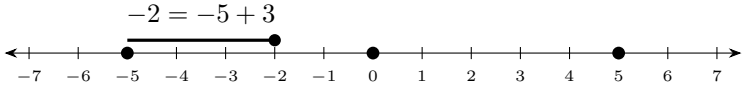
Since $2k^2 + 2k \in \mathbb{Z}$ by closure, n^2 has the form $2m + 1$ for an integer m. Therefore n^2 is odd. As n was arbitrary, the claim follows. ■

5 Quotient–remainder theorem and floor

For every integer a and positive integer d, there exist *unique* integers q, r such that

$$a = dq + r, \qquad 0 \le r < d.$$

Here q is the quotient and r is the remainder. In fact $q = \lfloor a/d \rfloor$. For $a = 17, d = 5$, we get $17 = 5(3) + 2$. For $a = -17, d = 5$, we get $-17 = 5(-4) + 3$; the remainder is 3, *not* -2, because $0 \le r < 5$.



Worked example: find the quotient and remainder

Question. Write $-23 = 7q + r$ with $0 \le r < 7$.

Solution. Since $\lfloor -23/7 \rfloor = -4$, choose $q = -4$. Then $r = -23 - 7(-4) = 5$. Thus $-23 = 7(-4) + 5$, and $0 \le 5 < 7$.

6 Primes, composites, and prime factorization

A **prime** is an integer $p > 1$ whose only positive divisors are 1 and p. A **composite** is an integer $n > 1$ with a positive divisor strictly between 1 and n. The numbers 0 and 1 are neither prime nor composite.

Worked example: factoring a number into primes

Question. Find the prime factorization of 756.

Solution. Divide repeatedly by primes: $756 = 2 \cdot 378 = 2^2 \cdot 189 = 2^2 \cdot 3 \cdot 63 = 2^2 \cdot 3^3 \cdot 7$. Thus $\boxed{756 = 2^2 \cdot 3^3 \cdot 7}$.

$$\begin{gathered} 756 = 2 \times 378 = 2 \times 2 \times 189 \\ 189 = 3 \times 63 = 3 \times 3 \times 21 = 3 \times 3 \times 3 \times 7 \\ \therefore\ 756 = 2^2 \cdot 3^3 \cdot 7 \end{gathered}$$

The **Fundamental Theorem of Arithmetic** says every integer greater than 1 has a prime factorization unique up to ordering. This allows us to compare divisibility by comparing prime exponents.

7 Greatest common divisor and least common multiple

The greatest common divisor $\gcd(a, b)$ is the greatest positive integer dividing both a and b (for integers not both zero). The least common multiple $\operatorname{lcm}(a, b)$ is the least positive integer divisible by both nonzero integers in absolute value.

Worked example: compute gcd and lcm from prime exponents

Question. Find $\gcd(72, 120)$ and $\operatorname{lcm}(72, 120)$.

Solution. We have $72 = 2^3 \cdot 3^2$ and $120 = 2^3 \cdot 3 \cdot 5$. For the gcd take the *minimum* exponent of each prime: $2^3 \cdot 3 = 24$. For the lcm take the *maximum*: $2^3 \cdot 3^2 \cdot 5 = 360$.

8 Euclid's lemma and proof by contraposition

Euclid's lemma: if p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$. The primality assumption matters: $6 \mid (2 \cdot 3)$, yet $6 \nmid 2$ and $6 \nmid 3$.

Worked example: an even square implies an even number

Question. For every integer n, if n^2 is even, then n is even.

Proof. Let $n \in \mathbb{Z}$ be arbitrary. We prove the contrapositive. Suppose n is odd. Then $n = 2k + 1$ for some $k \in \mathbb{Z}$, so

$$n^2 = 2(2k^2 + 2k) + 1.$$

Because $2k^2 + 2k \in \mathbb{Z}$, the square is odd. This establishes the contrapositive, hence the original implication. Since n was arbitrary, the claim follows. ■

9 Rational and irrational numbers in proofs

A real number is rational if it equals a/b for integers a, b with $b \neq 0$. A complete proof must verify that the numerator and denominator remain integers *and* that the new denominator is nonzero.

Worked example: adding a rational to an irrational

Question. If x is irrational and r is rational, then $x + r$ is irrational.

Proof. Let x be irrational and r rational. Suppose, for contradiction, that $x + r$ is rational. Then $x + r = a/b$ and $r = c/d$ for integers a, b, c, d with $b, d \neq 0$. Subtracting gives

$$x = \frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}.$$

The numerator and denominator are integers by closure. Moreover $bd \neq 0$ because $b, d \neq 0$. Thus x is rational, contradicting our assumption. Therefore $x + r$ is irrational. ■

10 Common proof-writing mistakes

- Reversing the notation $a \mid b$ and $b \mid a$.
- Using $2k$ or $2k + 1$ without specifying that k is an integer.
- Forgetting the closure statement after factoring.
- Giving a negative remainder when the divisor is positive.
- Applying Euclid's lemma to a composite divisor.
- Ending a proof before stating why the definition has been satisfied.

11 Practice questions

1. Prove that if $4 \mid a$ and $4 \mid b$, then $4 \mid (5a - 2b)$.
2. Prove that the product of an even integer and an odd integer is even.
3. Write $-41 = 6q + r$ with $0 \leq r < 6$.

4. Find $\gcd(84, 126)$ and $\text{lcm}(84, 126)$.
5. Disprove: if $12 \mid ab$, then $12 \mid a$ or $12 \mid b$.
6. Prove that for every integer n, $n(n+1)$ is even.

12 Fully worked practice solutions

1. **Proof.** Let $a, b \in \mathbb{Z}$ be arbitrary with $4 \mid a$ and $4 \mid b$. Then $a = 4r, b = 4s$ for some integers r, s. Thus $5a - 2b = 4(5r - 2s)$. By closure $5r - 2s \in \mathbb{Z}$, so $4 \mid (5a - 2b)$. ■
2. **Proof.** Let $a = 2r$ be even and $b = 2s + 1$ odd, where $r, s \in \mathbb{Z}$. Then $ab = 2[r(2s + 1)]$. Since $r(2s + 1) \in \mathbb{Z}$, ab is even. ■
3. $q = \lfloor -41/6 \rfloor = -7$ and $r = -41 - 6(-7) = 1$. Thus $-41 = 6(-7) + 1$.
4. $84 = 2^2 \cdot 3 \cdot 7$ and $126 = 2 \cdot 3^2 \cdot 7$. Hence gcd $= 2 \cdot 3 \cdot 7 = 42$ and lcm $= 2^2 \cdot 3^2 \cdot 7 = 252$.
5. Choose $a = 3, b = 4$. Then $12 \mid ab$ since $ab = 12$, but $12 \nmid 3$ and $12 \nmid 4$. Therefore the universal claim is false.
6. **Proof by cases.** Let $n \in \mathbb{Z}$ be arbitrary. If n is even, write $n = 2k$, so $n(n + 1) = 2[k(n + 1)]$, even by closure. If n is odd, write $n = 2k + 1$, so $n + 1 = 2(k + 1)$ and $n(n + 1) = 2[n(k + 1)]$, even by closure. These two cases exhaust all integers. Hence $n(n + 1)$ is always even. ■